AI Project Proposal

Trading Card Game (TCG) Card Recognizer

1. Project Overview

Trading card games such as Pokémon, Magic: The Gathering, and Yu-Gi-Oh! have millions of active collectors worldwide. Identifying cards, determining their edition, rarity, and market-relevant attributes is a time-consuming manual process that requires expert knowledge. Mistakes lead to mispriced trades and missed value.

This project builds an **AI-powered card recognition system** that lets a user photograph (or upload an image of) any supported TCG card and instantly receive a full classification of its attributes. For a Pokémon card, for example, the system will identify the Pokémon name, type (Fire, Water, …), level, HP, collection/set, card number, rarity, and edition. The recognised card data is then enriched through publicly available TCG APIs (e.g. Pokémon TCG API, Scryfall) to return complete metadata.

The target users are TCG collectors, casual players, and small-scale resellers who need fast and reliable card identification without specialised knowledge.

2. MVP Goals (Minimum Viable Product)

Included in the MVP:

- Image capture or upload of a single TCG card (starting with Pokémon).
- A custom-trained image recognition model that identifies the card from a self-built image dataset, supporting cards in multiple orientations and lighting conditions.
- Multi-label classification returning key attributes: name, type, collection/set, rarity, and card number.
- Enrichment of predictions via the Pokémon TCG API to fill in remaining metadata (HP, attacks, market price indicator).
- A simple web interface where the user uploads an image and receives the full card profile.

Not included yet:

- Support for other TCGs beyond Pokémon (Magic, Yu-Gi-Oh!, etc.).
- Real-time camera detection and batch scanning of multiple cards.
- Price tracking, collection management, or marketplace integration.

3. AI Technologies Used

- Computer Vision – The foundation of the project. A Convolutional Neural Network (CNN) processes card images and extracts visual features (artwork, layout, symbols) for identification.
- Machine Learning (multi-label classification) – A classification head on top of the CNN maps extracted features to multiple attribute labels simultaneously (type, rarity, set, etc.).
- Transfer Learning – We will fine-tune a pre-trained model (e.g. ResNet or EfficientNet) on our custom card dataset rather than training from scratch, which drastically reduces the amount of training data and time required.
- Data Augmentation – Cards will be photographed and augmented (rotation, skew, brightness, blur) to ensure the model generalises to real-world conditions such as varying angles and lighting.
- API Integration (rule-based enrichment) – Model predictions are cross-referenced with public TCG APIs to validate and complete the card profile, combining AI outputs with deterministic data look-ups.

These techniques are chosen because the core challenge is a visual classification task: the system must learn to map pixel data to a structured set of labels. A CNN handles the image understanding, while the classification head maps features to the many attribute categories. Combining model predictions with API look-ups keeps the architecture simple yet accurate.